Telco Customer Churn Prediction Report

Approach

The Telco Customer Churn dataset was loaded and explored using pandas. The dataset contains 7,043 customer records and 21 columns. The target variable was Churn, which indicates whether a customer left the company.

The data was checked for missing values. Blank values in the TotalCharges column were converted to missing values and filled using the median. Categorical variables were converted into numeric values using one-hot encoding. The customerID column was removed because it is only an identifier and does not provide useful predictive information.

The data was divided into 80 percent training data and 20 percent testing data. Stratification was used to preserve the proportion of customers who churned and did not churn. The features were standardized before training the KNN model.

Model Performance

Several KNN models were tested using K values of 1, 3, 5, 7, 9, 11, and 15. The selected model used K equal to 11. It achieved an accuracy of 77.36%, a precision of 57.70%, and a recall of 55.08%.

For comparison, the K=5 model achieved 74.73% accuracy, 52.53% precision, and 50.00% recall. The K=11 model performed best overall among the tested values, particularly because it achieved the highest recall and accuracy.

Accuracy measures the overall percentage of correct predictions. Precision measures how many customers predicted to churn actually churned. Recall measures how many of the customers who actually churned were correctly identified.

Key Findings

The analysis suggests that customers with month-to-month contracts, shorter tenure, higher monthly charges, and certain internet service or payment methods may have higher churn rates. These features appear to be useful indicators of customer churn.

Business Recommendations

The company should use the model to identify customers who may be at risk of leaving. These customers could receive personalized retention offers, improved technical support, or discounts.

The company should pay particular attention to customers with month-to-month contracts and short tenures. It may also be useful to encourage customers to switch to longer-term contracts.

Recall should be monitored carefully because failing to identify a customer who churns may result in lost revenue. However, precision should also be considered because contacting too many customers who would not churn could increase marketing costs.

Limitations

The KNN model is sensitive to feature scaling, although standardization was applied. The dataset also contains more customers who stayed than customers who churned, so accuracy alone may not fully describe the model's performance.

The model was evaluated using one train-test split rather than cross-validation. KNN does not provide direct feature-importance values, so feature patterns were explored using churn rates and group comparisons.

The dataset represents historical customer behavior, so future customer behavior may be different. Future improvements could include cross-validation, tuning additional model parameters, and comparing KNN with models such as logistic regression, decision trees, or random forests.